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GET 102 — Past Questions & Materials  
**by Renance**

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Course: GET 102  
Institution: Federal University of Petroleum Resources, Effurun  
Papers in this pack: 2  
Levels: 100

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# Paper 1 of 2 — GET 102 ENGINEERING GRAPHICS AND SOLID MODELLING 2023/2025

Level 100 • First Semester • PastQuestion

UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

DEPARTMENT OF MECHANICAL ENGINEERING

SECOND SEMESTER EXAMINATION 2023/2024 ACADEMIC SESSION

COURSE CODE AND TITLE: GET 102- ENGINEERING GRAPHICS AND SOLID MODELLING

TIME ALLOTTED: 3 HOURS

INSTRUCTIONS: Answer all Questions. All drawings must be neatly dimensioned. Construction lines should be clearly shown. All dimensions are in millimeters (mm).

Bolderline and Titleblock =10 Marks.

## QUESTION 1

- (a) Make an isometric drawing of the object shown in figure Q1a in 1:1,
- (b) Draw the third angle projection of the object in figure Q1a, from the direction of arrow X. 15 marks

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Figure Q1a.

Figure Q2.

## QUESTION 2

A first angle orthographic views of an object is shown in figure Q2, draw the isometric projection in 2:1

10 marks

## QUESTION 3

- (a) Inscribe and circumscribe a circle in a single triangle ABC, where BC (Baseline) = 120mm,  $\angle ABC$  and  $\angle ACB = 60^\circ$

- (b) Using a compass, inscribe a hexagon in a circle of radius 60 mm.

- (c) Using the concentric method, construct an ellipse with major axis AB=100mm and minor axis CD=60mm

15 marks

## QUESTION 4 sig: SS,

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- (a) Briefly outline the steps you will take to set up a drawing space in AutoCAD.

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- (b) Taking the reference point (A) shown in figure Q2 as the origin P; (0,0). Make a freehand sketch of that view and write out the coordinate of all line-intersection points.
- (c) Write a sequence of command syntax of how to produce the same sketched view of question 4b in AutoCAD. 20 marks

## Paper 2 of 2 — GET 102 Engineering drawing.

Level 100 • Second Semester • Handout

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Manufacturing of a product is the main activity in engineering profession. The design of a product may start with trial designs in the form of sketches on paper. As the design improves and undergoes changes, the final form of design must be the scaled manufacturing drawings with finer details included. These drawings are two-dimensional representations of three-dimensional objects designed.

During the process of design, the designer may have to carry out a large amount of computations so that an optimum design is obtained. A computer with good graphic capabilities helps the designer to

- (i) realize his ideas.
- (ii) carry out complex computations.
- (iii) present the results of computations in a useful form for decision making and possible improvement.
- (iv) present the improved model for evaluation.

Interactive Computer Graphics (ICG) is the tool of the designer.

A part to be manufactured is defined first in terms of its geometry which also includes dimensions, tolerances, surface finish, and in some cases the type of fit between two mating parts. The two-dimensional representation of a part, called an engineering drawing or a blueprint, shows three orthogonal views of the part. Sometimes, when three views are not enough to define the part, additional sectional views, auxiliary views may have to be added for conveying the right information.

Any design is finally represented in the form of orthographic views and auxiliary views so that production can be carried out. Hence, the computer aided drafting is an important tool for Computer Aided Design. The Computer Aided Drafting (CAD) system is the computerization of technical, production, electronics and architectural drawings.

This chapter is written and contributed by Prof. Pramod R. Ingle of B and B Institute of Technology, Vallabh Vidyanagar.

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CAD is the product of computer era. Its development originated from early computer graphics systems. CAD can find its roots to the development of Interactive Computer Graphics (ICG). A system called Sketchpad was developed at Massachusetts Institute of Technology, U.S.A., in 1963 by Ivan Sutherland.

In the beginning, CAD systems were no more than graphics editors with some built-in design symbols. The geometry available to the user was limited to lines, circular arcs, and a combination of the two. The development of free-form curves and surfaces such as Coon's patch, Ferguson's patch, Bezier's curve, and 8-splines enabled a CAD system to be used for more sophisticated work. A 3D CAD system allows a user to do very sophisticated design and analysis work.

Computer aided drawing and drafting system uses the computer to assist in generation of blueprint data. CAD systems are essential in design and a large number of computer based systems are commercially available. 2D drawing systems correspond directly to traditional engineering drawings, and they are developed to substitute manual drafting. The advantages offered by computerized drafting systems can be summarized as:

- (a) It increases the accuracy and productivity of designer.
- (b) It allows design alterations to be made easily.
- (c) It offers better drawing visualization through colours.
- (d) It improves the quality of drawings produced.
- (e) Drawings are easier to store and retrieve.
- (f) Storage space required is less.
- (g) Transfer of drawings is faster and cheaper.
- (h) It permits the use of library of standard symbols for more productive CAD work (refer to art. 26-6).

Although the capital investment required in setting up a computer aided drafting system is high, the greater capabilities offered by computers and software are making the systems more affordable.

A computer system consists of

- (a) Central Processing Unit (CPU), also known as processor
- (b) Main memory
- (c) Input devices
- (d) Output devices
- (e) Secondary storage devices.

The CPU controls and supervises the entire computer system. The actual arithmetic and logical operations are performed by CPU with the help of main memory. The main memory stores program instructions and processing data. Secondary storage devices provide slow but high capacity memory for storing program and large amount of data that is not currently being processed. Typical secondary storage devices are hard disk, floppy diskettes, magnetic tape, C.D. (compact disk), D.V.D., pen drive, etc.

During operation, data flows between the CPU, the main memory, and the various input/output devices. Moreover, the control information flows between these units to tell what to do with the data. Control information is used to manage the overall operation of the system.

Art. 26-3-2] Computer-Aided Drafting 623

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CONTROL UNIT -----,

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INPUT UNIT MEMORY UNIT

L ARITHMATIC - - .... Control Information

LOGIC UNIT -oataFlow

Organisation of a computer

FIG. 26-1

It must be remembered that the main memory is directly connected to the CPU.

The main memory stores the program instructions and all the data being used for processing until the data are released as output under the instructions from the CPU.

A control unit inside the CPU directs and coordinates all the operations of the system that are called for by the program. It involves control of the input and output devices, the storage and retrieval of information from memory, and the transfer of data between memory and ALU.

A CAD system consists of Hardware and Software. The system usually has the following major hardware elements:

Every CAD system must have a CPU to process and store a very large amount of graphics data. The speed and processing power of a computer is mainly decided by the CPU speed and the amount of main memory (RAM) installed.

The most common type of display device for CAD systems is a Cathode Ray Tube (CRT). The display device must be capable of displaying both graphical and alphanumeric data. Although the display looks like television monitors, they have special electronics for creating graphics images on the screen from computer processed data as per the instructions. But, they lack the television program receiving capability. Since both graphics and text are to be displayed simultaneously on the screen, the graphics monitor is usually divided into graphics area and a small text display area (see fig. 26-3).

The capabilities of display devices are evaluated in terms of resolution, colours and picture refresh rate. Picture refresh rate is important to get a steady and flicker free display. For a general purpose graphics system, a display with 1024 (horizontal) by 768 (vertical) pixel resolution, 0.28 mm dot pitch and 256 shades of colours is the desirable configuration. The small dot pitch gives a high resolution and therefore good quality picture.

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(1) [fig. 26-2(i)]: This is the most common input device. It is preferred for entering commands, text or value such as coordinates of a point or radius of a circle. The 101-key keyboards have special function keys to support special graphic functions. Whenever a key is pressed, the key character is identified by the computer and a character is displayed on the screen. If function keys are pressed or a combination of keys is pressed, the software takes the appropriate action.

(2) Mouse [fig. 26-2(iii)]: The mouse is a pointing device which is moved

across a flat surface (usually on a mouse pad) by hand to indicate X-Y movement. The rotation of rubber ball underneath the device is translated and the corresponding cursor movement on the screen provides a visual feedback. The mouse is used as a pointing and pick-up device.

Either two-button or three-button mice are available in the market. The mouse has become very popular with microcomputers in recent years, and it is an indispensable pointing device. It is inexpensive, small, and convenient to use. However, fine and very precise sketching is very difficult to obtain with the mouse.

(iv) DISPLAY

(iii) MOUSE

CAD system hardware

FIG. 26-2

(3) Tablet or Digitizer [fig. 26-2 (ii)]: This is also a very common and popular input device for CAD. It has a flat surface on which a stylus is placed and moved. The movement of stylus on the surface in terms of X and Y coordinates of surface is detected by sensors embedded in the tablet. The stylus incorporates a number of buttons for input. The parameters of a digitizer are resolution, accuracy and repeatability. The digitizers are available in very large sizes to be used with full size engineering drawings. They have a very high resolution. The resolution is measured in terms of lines per inch (LPI). Digitizers with 2000 LPI resolution are commercially available. The most common method of using digitizer is for tracing a drawing or a portion of the drawing. The drawing to be copied is placed on the digitizer and the stylus is used to trace the portions required. Thus, old drawings, maps, etc., can be entered into a CAD system.

Art. 26-4] Computer-Aided Drafting 625

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A hard copy i.e. copy on paper of what is displayed on a graphics terminal can be obtained by a variety of graphics printers and plotters. The computer graphic output device may be thought of as paper and a pen or a pencil.

(1) Dot Matrix Printers (DMP) and laser Printers: Dot Matrix Printer is the most commonly used printer for text printing. The characters are formed by printing dots in a specific manner. The dot matrix printer [fig. 26-2(vi)] can also be used for printing of drawings, but the quality of output is poor.

(2) Pen Plotters [fig. 26-2(v)]: Pen plotters are the simplest output devices for CAD. A pen plotter consists of a device to hold the paper. Usually two orthogonal motorized carriages hold a pen and move it under computer control. There are three inputs to the pen plotter: (i) an X coordinate, (ii) a Y coordinate and (iii) a pen variable. The pen variable can specify the pen colour by pen number, the pen to be up (non-drawing position) or down (in contact with paper in drawing position).

The two varieties of plotters are flat-bed and drum plotters. The flat-bed plotter is limited by the paper size it can handle. The drum plotter utilizes a continuous roll of paper which rolls over the top of the drum. The capacity and capabilities of a plotter are evaluated by the size of paper it can handle, resolution, speed of plotting, number of pens it can handle, etc.

(3) Ink-Jet printers/plotters: These are dot matrix printers. The drawing which is made up of lines, arcs, characters and symbols is converted into dot form. Then the rows of dots are printed across the width of the paper by impelling a tiny jet of ink on the surface of paper. The jets are switched on and off at high frequency to create multicolour plots. Typically, the resolution is 600 to 900 dots per inch, and each dot is arranged to overlap the adjacent ones. This provides a high quality photo-realistic picture. These plotters are quiet during operation. These plotters are used for colour plots of drawings, shaded images, contour plots and artistic work.

(4) Laser printers: A laser printer rapidly produces high quality text and graphics on plain paper. The fastest colour laser printers can print over 100 pages per minute. It can write with much greater speed than an ink jet and can draw more precisely, without spilling any excess ink. The toner powder of laser printer is cheap and lasts a long time compared to expensive ink cartridges.

The CAD system creates an environment to prepare drawings interactively. Most CAD systems available commercially are menu driven. Commands can either be typed directly with the help of a keyboard or can be picked-up from the screen menu or from toolbar with the help of a mouse or can be selected from the digitizer menu. Some screen menus offer pull-down menus (also referred to as pop-up menus) and dialogue boxes. For example, a variety of hatching patterns are displayed on the screen for better visualization and selection if a hatching command is chosen. The appropriate hatching pattern can be selected with the help of input device. The effect of every command is immediately displayed on the screen so that selection and corrections can be done interactively and immediately.

#### 626 Engineering Drawing [Ch. 26]

The major functions to be performed by a computer aided drafting system are:

- (a) Basic set-up of a drawing
- (b) Drawing the objects
- (c) Changing the object properties
- (d) Translating the objects
- (e) Scaling the objects
- (f) Clipping the objects to fit the image to the screen
- (g) Creating symbol libraries for frequently used objects
- (h) Text insertion
- (i) Dimensioning
- (j) Creates various layers (Transparent sheets)
- (k) Allows zoom-in and zoom-out of any components of drawing or complete

drawing

(I) Creates different numbers of print/plot layouts.

Some of the features of CAD systems are:

- (1) Modelling and Drafting: The majority of systems provide 2D and 3D modelling capabilities. Some low cost CAD systems are dedicated to 2D drafting only.
  - (2) Ease of use: The users find CAD systems very easy to learn and use.
  - (3) flexibility: Popular CAD systems provide greater flexibility when configuring the available hardware. Hundreds of computers, display devices, expansion boards, input and output devices are compatible and configurable with popular softwares.
  - (4) Modularity: Standard input and output devices are attached to standard connectors thereby making the system modular in nature.
  - (5) low maintenance cost: Little maintenance is needed to keep the system functional.
- Capabilities and versatilities of the drafting system vary depending on the system on which they are implemented. AutoCAD, VersaCAD, CADKey, DesignCAD, ZWCAD, etc. are few popular commercially available drafting systems in use. These systems provide a variety of features required for producing engineering drawings.

As an example of a popular, low-cost CAD software, we describe here some of the fundamental capabilities of AutoCAD. All these or similar facilities are very much desired in any CAD software for it to be useful.

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AutoCAD (a product and registered trademark of Autodesk Inc., USA) is a low cost yet very effective computer aided design and drafting software. AutoCAD is accepted as the industry standard and it is preferred by a large community of CAD users in the world. Although AutoCAD is available for a variety of computer systems, majority of AutoCAD implementations are available on IBM or compatible personal computers with various operating systems. AutoCAD comes with a very large number of user-selectable options to support a great variety of commercially available display devices, digitizers, mouse, printers and plotters. AutoCAD supports 2D drafting and 3D modelling.

Art. 26-5-2] Computer-Aided Drafting 627

The basic drawing entities are lines, polylines [refer to art. 26-5-4 of any width, circles, arcs, ellipses and solids. There are many ways of defining a drawing entity, and the software always prompts the user for all options. Each drawing entity has an associated line-type, colour, layer [refer to art. 26-5-5(2)] and thickness. The thickness is a property associated with 3D entities.

Before any drawing is started, the AutoCAD environment must be prepared for proper units of measurement, line-type, drawing size, layer, etc. In AutoCAD the drawings are always prepared at full scale, and the drawing size can be changed at any instant of time by using LIMITS command.

- (1) P IV Intel processor and motherboard
- (2) 160/320 GB Hard disk
- (3) 1 GB/2 GB RAM

(4) Microsoft Window XP Operating system software.

The default AutoCAD 2010 drawing screen can be divided into four areas such as drawing area, command area, menu bar area and tool bar area. The various component of Graphical User Interface (GUI) of AutoCAD 2010 is shown in fig. 26-3.

Properties tool bar

Draw toolbar

Modify tool bar

WCSicon

Command

prompt window

display Graphic cursor

+1

POLAR Switch Scroll bar -----)o-

Status bar

t

Components of AutoCAD Classic screen of AutoCAD 2010

FIG. 26-3

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F1 Online Help

F2 Toggles between Drawing screen to text screen

F3 Toggles between OSNAP On and Off

F4 Toggles between Tablet On and Off

F5 Switches among Isoplanes Top, Right and Left

F6 Toggles between Coordinates On and Off

F7 Toggles between Grid On and Off

F8 Toggles between Ortho Mode On and Off

F9 Toggles between Snap Mode On and Off

F10 Toggles between Polar Tracking On and Off

F11 Toggles between Objects Snap Tracking On and Off

(1) Line (Refer module 26-1 ): A line is specified by giving its two endpoints. The LINE command can be used to draw a single line or a series of lines with the end-point of one being the start-point of the next. When a series of such lines is created, each line is treated as a separate entity. To create a closed polygon, the user has to type in C (close option) for the To point: prompt. This causes the last and the first points to be joined by a line and thus creating a closed boundary.

Polyline (Refer module 26-9, 26-25): Polylines are interesting drawing entities. Polylines can include both lines and arcs connected at end-points. Thus, a polyline is a single entity with multiple

segments. The polylines can be straight or curved, can be wide (like a TRACE) or tapered. Fillets and chamfers can be added where needed on a polyline. Curve fitting and hatching can easily be performed on a polyline.

(fig. 26-4) (Refer module 26-8, 26-12): A polygon

is also a polyline with equal length of sides. The regular polygon can either be inscribed in a circle or circumscribed about the circle. The polygon may also be constructed by specifying the length of one side and the number of sides of polygon (called edges). In this method a polygon is constructed in anti-clockwise direction from the two edge end-points that have been specified.

Art. 26-5-4] Computer-Aided Drafting 629

#### INSCRIBED CIRCUMSCRIBED EDGE

Polygon

FIG. 26-4

(5) Arc (fig. 26-5) (Refer module 26-8, 26-26, 26-27): This command is used to draw an arc accurately. Usually there are three parameters required for drawing an arc.

Different ways of drawing circular arcs are:

- (i) 3 point arc: The arc is drawn by specifying three points on the chord of arc. The first and third points define the start and end-points of an arc respectively.
- (ii) Start, Center: This option needs start point and center point of an arc. The third parameter may either be an end-point, included angle, or length of chord.
- (iii) Start, End: This option asks the user to enter the start and end-points of an arc. The arc is completed by either specifying radius or included angle or center point.

S.P.

Second

Point

E.P.

(1) 3-point arc: Start point, Second point, (3) Center point, Start point, End point (2) Center point, Start point, End point Included angle(e)

r Length of chord

E.P.

R

Xe S.P.

S.P " ' - - - -

(4) Center point, Start point (6) Start point, End point, Length of chord (5) Start point, End point, Radius Included angle(e)

Arc definitions

FIG. 26-5

## 630 Engineering Drawing [Ch. 26]

(6) Circle (Refer module 26-5, 26-11, 26-'13): There are many ways of drawing a circle, the default being the centre point of circle and radius. Either on typing the command CIRCLE or selecting it from a menu bar with the help of mouse, all circle drawing options are displayed. The options available are:

(a) Center point and Radius

(b) Center point and Diameter

(c) 3P: This specifies 3 points on the circumference of a circle. There is a unique circle passing through three given non-collinear points.

(d) 2P: This specifies the end-points of diameter of a circle.

(e) TTR (Tangent Tangent Radius): This command draws a circle of specified radius that is tangent to two lines, circles or arcs.

(f) TTT (Tangent Tangent Tangent): This command draws a circle tangent to three entities.

(7) Ellipse (Refer module 26-11 ): An ellipse can be constructed by specifying the center point, radial length of major and minor axes. An ellipse can also be constructed by specifying end-points of one of its axis and the radial length of other axis.

(8) Donut (fig. 26-6) (Refer module 26-10): The DONUT is a special type of polyline which is made up of arc segments. A DONUT has two properties: it has width, and it is closed. The width of DONUT is set by specifying inside and outside diameters. The inside diameter may be zero thereby making it a filled circle.

(9) Hatch patterns (Refer module 26-12): The HATCH command Donut examples is used to fill up the area using a suitable pattern. The type of Fig. 26-6 pattern and pattern variables can be chosen from a library of patterns available. The hatching will be carried out inside a closed defined area.

(10) Text (Refer module 26-6): Words, messages and numbers can be inserted as required on an engineering drawing. The alphanumeric keyboard is used extensively for non-graphical input such as text. The text style, height, text angle, aspect ratio, colour, etc. are some of the attributes associated with text. These attributes can be changed as per requirements.

(11) Rectangle (Refer module 26-6, 26-10): Rectangles are drawn by three methods

(i) By specifying the co-ordinates at the "specify first corner point" prompt and at the "specify other corner point" prompt.

(ii) By entering area of rectangle in current units and specifying length or width of rectangle. For this select "Area" option of rectangle command.

(iii) By specifying length distance and width distance of rectangle. For this select "Dimensions" option of rectangle command.

(1) limits (Refer module 26-1 to 26-13): In AutoCAD, drawings are drawn in full ,~4'

scale therefore limits are needed to size up a drawing area. The limits are set by specifying X and Y co-ordinates of Lower Left Corner (L.L.C.) and Upper Right Corner

(U.R.C.) of drawing area. By default L.L.C. is set to (0, 0) and U.R.C. is set to (12, 9) for imperial file and (420, 210) for metric file.

Note: By changing limits, the drawing display area does not change. It is required to use all option of the zoom command to display the limits inside the drawing area.

Art. 26-5-5] Computer-Aided Drafting 631

(2) layer (Refer module 26-6, 26-13): A layer can be thought of as a transparent sheet on which drawings can be prepared. Drawings can be logically divided into different layers, and layers can be selectively displayed either individually or in any combination. Each layer is identified by a name. If the drawing becomes too dense or complicated, some layers can be turned off so that they do not interfere with the work. The drawing can be edited on any one layer at a time, called current layer. The colour, linetype and line weight are the properties associated with a layer. This controls an entity's colour, linetype and line weight drawn on that layer.

(3) Dimensioning (Refer module 26-7): The manufacturing drawing must be dimensioned for size and tolerances so that the right information can be conveyed. The appearance and size of dimension arrows, size and style of dimension text with or without tolerances, and the layer on which dimensions are placed can be controlled by setting dimension variables.

(4) Object snap (fig. 26-7) (Refer module 26-4): A very useful drawing aid, the OSNAP identifies the points on drawing entities that are visible on the screen. For example, the start point of an arc can be the endpoint of an existing line. This option allows the user to pick-up the points very accurately with respect to drawing displayed. Some of the OSNAP modes are:

z NEArest Closest point on an entity.

Center of a circle or an arc.

0 CENTER

.6. MIDpoint Midpoint of a line or an arc.

D ENDpoint Closest endpoint of a line or an arc.

X INTERsection Intersection of two lines, two arcs, or a line with an arc or a circle.

D TANgent

GJPI~.~~r

Tangent to an arc or circle.

<) Quadrant Quadrant points will be located for circle and ellipses.

CENTER MIDpoint ENDpoint

INTERsection TANgent QUAdrant

OSNAP options

FIG. 26-7

(5) Zoom (Refer module 26-1 ): This is the most common method of magnifying a portion of current drawing on the screen. The portion of drawing to be zoomed is

usually selected by a window. A window is identified by picking up the diagonal corners of a rectangle around the area of interest. Zoom also helps improving the accuracy of

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632 Engineering Drawing [Ch. 26

Modify Toolbar

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ORIGINAL I MIRROR IMAGE

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(iii) COPY (iv) MIRROR

Common drawing editing commands

FIG. 26-8

• The fundamental commands to edit a drawing are:

(1) Move [fig. 26-8(i)]: Moves selected objects to another location about a base point.

(2) Rotate [fig. 26-8(ii)] (Refer module 26-'12): Rotates selected objects through a specified angle about a base point.

(3) Copy [fig. 26-8(iii)] (Refer module 26-6): Creates one or more copies of selected

objects at another location. The function of COPY command is similar to the MOVE command except that it preserves a copy of the objects selected at the original location.

(4) Mirror [fig. 26-8(iv)] (Refer module 26-8): Creates a mirror image of the selected objects about a specified line.

(5) Array (Refer module 26-10, 26-11 ): This command creates multiple copies of selected objects in rectangular or polar form. This is a form of COPY command.

(6) Erase (Refer module 26-8): This command deletes the selected entities. A record of entities erased is always maintained. The most recent entity can be unerased by OOPS command.

(7) Oops (Refer module 26-14 ): This command retrieves all objects erased by the last Erase and after executing Block or Wblock command.

(8) Break: This command erases a portion of line, arc, circle or a 2D polyline between two selected points.

#### Art. 26-5-6] Computer-Aided Drafting 633

(9) fillet (Refer module 26-5, 26-13): This command is used to create a round corner between two lines. The lines are shortened or extended to fit a tangent arc of specified radius. FILLET works on any combination of two lines, arcs, circles, non-parallel lines, or a single polyline.

(10) Chamfer (Refer module 26-8): This command works on two lines or a single polyline to create a bevelled edge.

(11) Extend (Refer module 26-11 ): This command extends the lines, polylines and arcs to a boundary edge which can be a line, polyline, arc or circle. A closed polygon cannot be extended. When you invoke this command, you will be prompted to select the boundary edges. These edges can be lines, polylines, circles, arcs, ellipse, xlines, rays, splines etc. After the boundary edges are selected, you must select each object to be extended. An object can be both a boundary edge and an object to be extended.

(12) Offset (Refer module 26-5): This command creates a parallel single copy of line, arc, circle, rectangle, polygon, or 2D polyline at a given offset distance. Each offset creates a new entity with the same linetype, color and layer settings.

(13) Stretch (Refer module 26-12): The STRETCH command can either lengthen entities or shorten them, and thus alter their shapes. The centre points of arcs or polyline arcs are adjusted accordingly.

(14) Trim (Refer module 26-5, 26-13): This command trims the objects that extend beyond a required point of intersection. When you invoke this command, you will be prompted to select

the cutting edges. These edges can be lines, polylines, circles, arcs, ellipse, xlines, rays, splines, text, blocks or even viewports. After the cutting edges are selected, you must select each object to be trimmed. An object can be both a cutting edge and an object to be trimmed.

(15) Scale (Refer module 26-12): The SCALE command allows to shrink or enlarge the already existing drawing objects about a base point by specifying a scale factor.

(16) Pedit: A polyline is a single entity which is made up of a continuous series of line and arc segments. The PEDIT command is exclusively used for editing of polyline properties.

The selected line, arc and polyline can be added to an existing polyline by a JOIN option. A smooth curve passing through all vertices of a polyline can be created by using FIT option. Similarly, a spline can also be constructed by using SPLINE option.

(17) Explode (Refer module 26-6): This command breaks a polyline into its individual segments. These segments can then individually be edited, and rejoined again to form an edited polyline.

(18) U (Refer module 26-8): The U command reverses the effects of a series of previously used commands and hence allows back-stepping. The REDO command wipes out the effect of U command.

The advanced features of AutoCAD are:

(1) Built-in programming language AutoLISP provides programming environment so that AutoCAD commands can be called along with programs written for computations. This is very useful for parametric design and drawing.

(2) Drawing Exchange Files (DXF) and script files can be used to interface AutoCAD with programs written in any other higher level language such as C. DXF and IGES file formats allow the exchange of drawing files among various drafting softwares.

(3) AutoCAD provides techniques to define and extract attributes of entities. This feature is used in extracting information from a drawing for processing by other programs or to transfer it to a database.

Note: The commands not explained here above are covered up in the topics 26-6, 26-7, 26-8 and 26-9 as illustrative modules with on hand explanation.

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~4

A complete drawing or a part of a drawing can be given a specific name and then stored as a block. A collection of such blocks form a library which is very useful for drawing purpose. The blocks, also referred to as symbols, can be scaled, rotated or mirrored if necessary and inserted into the drawing at the appropriate location e.g. bolt heads to be used at different locations in a drawing. Many suppliers of CAD add-on products offer symbol libraries of standard mechanical components, electronic components, architectural symbols, piping symbols, etc. The block and wblock command groups a number of selected entities together and treats them as a single object i.e. single block. Block commands saves the group of selected entities with block name and is inserted in the file in which it is created, but wblock command saves the group of selected entities as a drawing file and can be inserted in any drawing file. The blocks and wblocks can be scaled, rotated, stretched or mirrored. The explode command separates the entities of block and wblock.

INSERT (Refer module 26-14): INSERT command is used to insert blocks and wblocks in current drawing. Blocks will be inserted only in the drawing file in which they created and wblocks will be inserted in any drawing files.

.,,.,,.

..r~

The following modules (26-1 to 26-14) are self interactive and self learning practice modules to generate two dimensional drawings covering all possible 2-D commands. The configuration of computer will be as mentioned in art. 26-5-1.

(1) Absolute coordinate method: In the absolute coordinate method, the points are located with respect to the origin (0, 0). The syntax is X, Y.

Module 26-1. Draw a line diagram as shown in fig. 26-9. Use Absolute Coordinate Method.

To draw a line diagram using Absolute Coordinate Method, follow the steps mentioned below.

(1) Command: LIMITS (.J)

Reset Model space limits:

Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (.J)

Specify upper right corner <12.0000,9.0000>: 120,90 (.J)

(2) Command: ZOOM (.J)

Specify corner of window, enter a scale factor (nX or nXP), or

[All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>: ALL (.J)

Regenerating model.

(3) Command: LINE (.J)

Specify first point: 20,20 (.J)

Specify next point or [Undo]: 100,20 (.J)

Specify next point or [Undo]: 100,50 (.J)

Specify next point or [Close/Undo]: 90,50 (.J)

Specify next point or [Close/Undo]: 85,65 (.J)

Specify next point or [Close/Undo]: 75,65 (.J)

Specify next point or [Close/Undo]: 70,50 (.J)

Specify next point or [Close/Undo]: 50,50 (.J)

Specify next point or [Close/Undo]: 50,65 (.J)

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Specify next point or [Close/Undo]: 30,65 (..)J

Specify next point or [Close/Undo]: 30,50 (..)J

Specify next point or [Close/Undo]: 20,50 (..)J

Specify next point or [Close/Undo]: C (..)J

(4) Save This File As Module 26-1.DWG

Output of Module 26-1 (fig. 26-9):

MODULE 26-1: ABSOLUTE COORDINATE METHOD

10 20 20 20 ,. I (

10 i, I

>I (

1<10>1( 20 3'/(

20 >I 5 I ( 10 >I 5 I ( 10 >I

Ps Ps

LO

P11 Ps P7 p4 p3

P12

LO

<O

80

50

70

75

85

90

100

ABSOLUTE COORDINATES OF POINTS P1 TO P12

P1 = 20 ,20 p4 = 90, 50 P7 = 70 ,50 P10= 30, 65

P2 = 100 ,20 Ps = 85 ,65 Ps = 50, 50 P11= 30, 50

P3=100,50 Ps = 75, 65 pg= 50 ,65 P12= 20,50

26-9 FIG.

(2) Relative coordinate method: In the relative coordinate method, the displacement along the X and Y axis are measured with reference to the previous point. The syntax is @X, Y.

Module 26-2. Draw a line diagram as shown in fig. 26- 10. Use Relative Coordinate Method.

To draw a line diagram using Relative Coordinate Method, follow the steps mentioned below.

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(1) Command: LIMITS (.J)

Reset Model space limits:

Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (.J)

Specify upper right corner <12.0000,9.0000>: 120,90 (.J)

(2) Command: ZOOM (.J)

Specify corner of window, enter a scale factor (nX or nXP), or

[All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>: All (.J)

Regenerating model.

(3) Command: LINE (.J)

Specify first point: 20,20 (.J)

Specify next point or [Undo]: @80,0 (.J)

Specify next point or [Undo]: @0,30 (.J)

Specify next point or [Close/Undo]: @-10,0 (.J)

Specify next point or [Close/Undo]: @-5,15 (.J)

Specify next point or [Close/Undo]: @-10,0 (.J)

Specify next point or [Close/Undo]: @-5,-15 (.J)

Specify next point or [Close/Undo]: @-20,0 (.J)  
 Specify next point or [Close/Undo]: @0,15 (.J)  
 Specify next point or [Close/Undo]: @-20,0 (.J)  
 Specify next point or [Close/Undo]: @0,-15 (.J)  
 Specify next point or [Close/Undo]: @-10,0 (.J)  
 Specify next point or [Close/Undo]: C (.J)

(4) Save This File As Module 26-2.DWG

Output of Module 26-2 (fig. 26-10):

MODULE 26-2: RELATIVE COORDINATE METHOD

10 20 20 20 10

I< >I < , I< , I< >I < 11-1

> I 5 I< 10 > I 5 I<

P10 pg P6 Ps

~1 P12 P11 Pa P7 p4 p3

RELATIVE COORDINATES OF POINTS P1 TO P12

P1 = 20,20 P4 = @-10,0 P1 = @-5,-15 P10 = @-20,0

P2 = @80,0 P5 = @-5,15 Pa = @-20,0 P11 = @0,-15

p3 = @0,30 P5 = @-10,0 Pg = @0,15 P12 = @-10,0

FIG. 26-10

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(3) Relative polar coordinate method: In the relative polar coordinate method a point is located by defining the direct distance of the point from current point and the angle that the direct distance makes with the positive X-axis in counter clockwise direction. It's syntax is @direct distance < angle.

Module 26-3. Draw a line diagram as shown in fig. 26-11. Use Relative Polar Coordinate Method.

Output of Module 26-3 (fig. 26-11):

MODULE 26-3. POLAR COORDINATE METHOD

1~J\_0\_.7~~~1+~~~-2\_0~-.,I>-j-, (~-2\_0~~~ ~~  
 2~

P10 \_\_\_\_~P9

f' P5 Ps

Pa j P7 \_120

I P1 ~-----~ P2

80

POLAR COORDINATES OF POINTS TOP12

P1 = 20, 20 P4 = @10 < 180 P7 = @15<252 P10 = @20 < 180

P2 = @80 < 0 Ps = @15 < 108 Pa= @20< 180 P11 = @14 .27 < 270

P3 = @30<90 Ps = @10<180 Pg= @14.27 < 90 P12 = @ 10.73 < 180

FIG. 26-11

To draw a line diagram using Polar Coordinate Method, follow the steps mentioned below.

(1) Command: LIMITS (.J)

Reset Model space limits:

Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (.4.J)

Specify upper right corner <12.0000,9.0000>: 120,90 (.J)

(2) Command: ZOOM (.J)

Specify corner of window, enter a scale factor (nX or nXP), or

[All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>: ALL (.J)

Regenerating model.

(3) Command: LINE (.J)

Specify first point: 20,20 (.J)

Specify next point or [Undo]: @80<0 (.J)

Specify next point or [Undo]: @30<90 (.J)

Specify next point or [Close/Undo]: @10<180 (.J)

Specify next point or [Close/Undo]: @15<108 (.J)

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Specify next point or [Close/Undo]: @10<180 (.J)

Specify next point or [Close/Undo]: @15<252 (.J)

Specify next point or [Close/Undo]: @20<180 (.J)

Specify next point or [Close/Undo]: @14.27<90 (.J)

Specify next point or [Close/Undo]: @20<180 (.J)

Specify next point or [Close/Undo]: @14.27 <270 (.J)

Specify next point or [Close/Undo]: @10.73<180 (.J)

Specify next point or [Close/Undo]: C (.J)

(4) Save This File As Module 26-3.DWG

Module 26-4. Draw a line diagram as shown in fig. 26- 12. Use Object-Snap (Osnap) Method.

To draw a line diagram using Object-Snap (Osnap) Method, follow the steps mentioned below.

(1) Command: LIMITS (.J)

Reset Model space limits:

Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (.J)

Specify upper right corner <12.0000,9.0000>: 120,90 (.J)

(2) Command: ZOOM (.J)

Specify corner of window, enter a scale factor (nX or nXP), or

[All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>: ALL (.J)

Regenerating model.

(3) Command: LINE (.J)

Specify first point: 20,20 (.J)

Specify next point or [Undo]: @80<0 (.J)

Specify next point or [Undo]: @45<90 (.J)

Specify next point or [Close/Undo]: @80<180 (.J)

Specify next point or [Close/Undo]: C (.J)

(4) Press F3 key to Switch ON "OSNAP" if it is OFF

(5) Command: LINE (.J)

Specify first point: (Get Mouse Near To P5 and CLICK When MIDPOINT magnet gets ON)

Specify next point or [Undo]: (Get Mouse Near To P6 and CLICK When MIDPOINT magnet gets ON)

Specify next point or [Undo]: (.J)

(6) Command: LINE (.J)

Specify first point: (Get Mouse Near To P7 and CLICK When MIDPOINT magnet gets ON)

Specify next point or [Undo]: (Get Mouse Near To P8 and CLICK When MIDPOINT magnet gets ON)

Specify next point or [Undo]: (.J)

(7) Command: LINE (.J)

Specify first point: (Get Mouse Near To P1 and CLICK When ENDPOINT magnet gets ON)

Specify next point or [Undo]: (Get Mouse Near To P3 and CLICK When ENDPOINT magnet gets ON)

Specify next point or [Undo]: (.J)

(8) Command: LINE (.J)

Specify first point: (Get Mouse Near To P2 and CLICK When ENDPOINT magnet gets ON)

Specify next point or [Undo]: (Get Mouse Near To P4 and CLICK When ENDPOINT magnet gets ON)

Specify next point or [Undo]: (.J)

(9) Save This File As Module 26-4.DWG

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Output of Module 26-4 (fig. 26-12):

MODULE 26-4: USE OF OBJECT-SNAP (OSNAP)

80

T

P71-----3>1E-----i Pa ~,

-----"P21

P6

USE ABSOLUTE, RELATIVE OR POLAR METHOD TO GET COORDINATES OF P1 TO P41

I REST OF THE POINTS SELECT BY OSNAP MAGNETS I

P1 = 20 ,20 Ps = ( Select Using MIDPOINT magnet) P1 = ( Select Using ENDPOINT magnet)

P2=@80<0 P6 = ( Select Using MIDPOINT magnet) p3 = ( Select Using ENDPOINT magnet)  
 p3 = @45<90 P7 = ( Select Using MIDPOINT magnet) P2 = ( Select Using ENDPOINT magnet)  
 p4 = @80< 180 Pa= ( Select Using MIDPOINT magnet) p4 = ( Select Using ENDPOINT magnet)  
 FIG. 26-12

Module 26-5. Draw a line diagram as shown in fig. 26-13. Use Circle, Offset, Trim and Fillet commands.

To draw a line diagram using circle, offset, trim and fillet commands, follow the steps mentioned below. After executing the commands in sequence, we will get the output as shown in fig. 26-13(i) to fig. 26-13(x).

(1) {Switch ON POLAR switch by Mouse Click or press F10 key}

(2) Command: LIMITS (.J)

Reset Model space limits:

Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (.J)

Specify upper right corner <12.0000,9.0000>: 120,90 (.J)

(3) Command: ZOOM (.J)

Specify corner of window, enter a scale factor (nX or nXP), or

[All/Center/Dynamic/Extents/Previous/Scale/Window] < real time>: ALL (.J)

(4) Command: LINE (.J)

Specify first point: 20,20 (.J) i ..-----.....

Specify next point or [Undo]: 90 (.J) (When 0° POLAR 201 ~ is ON)

20,2V 90 L1

Specify next point or [Undo]: 20 (.J) (When 90° POLAR is ON) FIG. 26-13(i)

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Specify next point or [Close/Undo]: 90 (.J) (When 180° POLAR is ON)

Specify next point or [Close/Undo]: C (.J)

(5) Command: OFFSET (.J)

Specify offset distance or [Through] <Through>: 35 (.J)

Select object to offset or <exit>: (Select Line L1)

Specify point on side to offset: (Click Mouse Above Line L1)

Select object to offset or <exit>: (.J) FIG. 26-13(ii)

(6) Command: CIRCLE (.J)

Specify center point for circle or [3P/2P/Ttr (tan tan radius)]: (Select MIDPOINT magnet of L3)

Specify radius of circle or [Diameter]: D (.J)

02 040

Specify diameter of circle: 40 (.J)

(7) Command: CIRCLE (.J) L3

Specify center point for circle or [3P/2P/Ttr (tan tan

radius]): (Select MIDPOINT magnet of L3)

Specify radius of circle or [Diameter] <20.0000>: D

(.J) L1

Specify diameter of circle <40.0000>: 20 (.J) FIG. 26--I 3(iii)

(8) Command: LINE (.J)

Specify first point:(Select 0° QUADRANT magnet Q1 of 040 Circle)

Specify next point or [Undo]:(Select PERPENDICULAR magnet on Line L2)

Specify next point or [Undo]: (.J)

(9) Command: LINE (.J)

Specify first point:(Select 180° QUADRANT magnet Q2 of 040 Circle)

Specify next point or [Undo]:(Select PERPENDICULAR magnet on Line L2)

Specify next point or [Undo]: (.J)

(10) Command: OFFSET (.J)

Specify offset distance or [Through] <35.0000>: 8

(.J) L2

L6

Select object to offset or <exit>: (Select Line L1)

~\_.\_,~~~"-L7

Specify point on side to offset: (Click Mouse Above

Line L1) A L14

Select object to offset or <exit>: (.J) FIG. 26-13(iv)

(11) Command: OFFSET (.J)

Specify offset distance or [Through] <8.0000>: 25 (.J)

Select object to offset or <exit>: (Select Line L6)

Specify point on side to offset: (Click Mouse on Right Side of Line L6)

Select object to offset or <exit>: (Select Line L7)

Specify point on side to offset: (Click Mouse on Left Side of Line L7)

Select object to offset or <exit>: (.J)

(12) Command: TRIM (.J)

Current settings: Projection=UCS, Edge=None L4 L2

Select cutting edges ... L6\_/ L7

Select objects: (Select Line L4) '--- L1

Select objects: (Select Line L5) FIG. 26-13(v)

Select objects: (.J)

Select object to trim or shift-select to extend or [Project/Edge/Undo]: (Select Line L2 from Circle Portion)

Select object to trim or shift-select to extend or [Project/Edge/Undo]: (.J)

Art. 26-7] Computer-Aided Drafting 641

(13) Command: TRIM (.J)

Current settings: Projection=UCS, Edge=None

Select cutting edges ... u o - ~ -L9

Select objects: (Select Line L9)

~L1

Select objects: (Select Line L10)

FIG. 26-13 (vi)

Select objects: (.J)

Select object to trim or shift-select to extend or [Project/

Edge/Undo]: (Select Line L1 from Arrow Head)

Select object to trim or shift-select to extend or [Project/Edge/Undo]: (.J)

(14) Command: FILLET (.J)

Current settings: Mode = TRIM, Radius = 0.0000

Select first object or [Polyline/Radiu?/Trim]: R (.J) L1o~L9

Specify fillet radius <0.0000>: 2.5 (.J)

L87

Select first object or [Polyline/Radius/Trim]: (Select Line FIG. 26-B(vii)

L8 from Arrow Head)

Select second object: (Select Line L10 from Arrow Head)

(15) Command: FILLET (.J)

Current settings: Mode = TRIM, Radius = 2.5000

Select first object or [Polyline/Radius/Trim]: (Select L9

L10

Line L8 from Arrow Head)

Select second object: (Select Line L9 from Arrow Head)

(16) Command: FILLET (.J)

Current settings: Mode = TRIM, Radius = 2.5000

Select first object or [Polyline/Radius/Trim]: R (.J)

Specify fillet radius <2.5000>: 5 (.J)

Select first object or [Polyline/Radius/Trim]: (Select L~L5

L11 ~ L2

Line L4 from Arrow Head)

Select second object: (Select Line L11 from Arrow L12 \_/ L13

Head) FIG. 2 6-13 (ix)

(1 7) Command: FILLET (.J)

Current settings: Mode = TRIM, Radius = 5.0000

Select first object or [Polyline/Radius/Trim]: (Select Line LS from Arrow Head)

Select second object: (Select Line L2 from Arrow Head)

(18) Command: FILLET (.J)

Current settings: Mode = TRIM, Radius = 5.0000

Select first object or [Polyline/Radius/Trim]: (Select

Line L11 from Arrow Head) L11 L2

Select second object: (Select Line L12 from Arrow

Head)

l13

(19) Command: FILLET (.J) FIG. 26-13(x)

Current settings: Mode = TRIM, Radius = 5.0000

Select first object or [Polyline/Radius/Trim]: (Select Line L2 from Arrow Head)

Select second object: (Select Line L13 from Arrow Head)

(20) Save This File As Module 26-5.DWG

Output of Module 26-5 (fig. 26-B):

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MODULE 26-5: USE OF CIRCLE, OFFSET, TRIM, AND FILLET COMMANDS

-----,-

RS

11~

RS

R2.5

l l

90

FIG. 26-13

Module 26-6. Draw a title block as shown in fig. 26-14. Use Rectangle, Explode, Text and Copy commands.

To draw a line diagram using Circle, Offset, Trim and Fillet commands, follow the steps mentioned below. After executing the commands in sequence, we will get the output as shown in fig. 26-14(i) to fig. 26-14(vii).

(1) Command: LIMITS (.J)

Reset Model space limits:

Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (.J)

Specify upper right corner <12.0000,9.0000>: 120,90 (.J)

(2) Command: ZOOM (.J)

Specify corner of window, enter a scale factor (nX or nXP), or

[All/Center/Dynamic/Extents/Previous/Scale/Window] < real time>: ALL (.J)

Regenerating model. REC-1

(3) Command: - LAYER (.J)

Enter an option

@100,45

[?/Make/Set/New/ON/OFF/Color/Ltype/LWeight/

... ]: NEW (.J)

Enter name list for new layer(s): GRID,TXT

(.J) 20,20

Enter an option

[?/Make/Set/New/ON/OFF/Color/Ltype/LWeight/ \_10\_0\_ \_ \_->,-a

... ]:COLOR (.J)

New color [Truecolor/Colorbook] : GREEN FIG. 26-'14(i)

(.J)

Enter name list of layer(s) for color 3 (green) <0>: GRID (.J)

Enter an option

[?/Make/Set/New/ON/OFF/Color/Ltype/LWeight/... J]:COLOR (.J)

Art. 26-7] Drafting 643

New color [Truecolor/Colorbook] : MAGENTA (.J)

Enter name list of layer(s) for color 6 (magenta) <O>: TXT (.J)

Enter an option

[?/Make/Set/New/ON/OFF/Color/Ltype/LWeight/ ... J]: (.J)

(4) Command: RECTANGLE (.J)

Specify first corner point or [Chamfer/Elevation/Fillet/Thickness/Width]: 20,20 (.J)

Specify other corner point or [Dimensions]: @100,45 (.J)

(5) Command: EXPLODE (.J)

Select objects: (Select Rectangle REC-1) /REC-1

10 10

Select objects: (.J) ~ - f - - l JI

(6) Use OFFSET and TRIM Command to

complete the Drawing as shown on

Right Side

(7) Command: -LAYER (.J)

Current layer: "O"

Enter an option

[?/Make/Set/New/ON/OFF/Color/Ltype/ FIG. 26-'1 4(ii)

LWeight ... J]: SET (.J)

Enter layer name to make current < 0 >: GRID (.J)

Enter an option

[?/Make/Set/New/ON/OFF/Color/Ltype/l Weight ... J]: (.J)

(8) Command: LINE (.J)

L1 86

Specify first point: (Select B1 by ENDPOINT r-87

81

Os nap) 82 -

Specify next point or [Undo]: (Select B6 83

84

by Osnap) 85-

Specify next point or [Undo]: (.J)

(9) Command: COPY (.J)

Select objects: (Select Line L1)

FIG. 26-14(iii)

Select objects: (.J)

Specify base point or disp... : (Select B1 by Osnap)  
 Specify second point of dis.. (Select B2 by Osnap)  
 Specify second point of dis.. (Select B3 by Osnap)  
 Specify second point of dis.. (Select B4 by Osnap)  
 Specify second point of dis.. (Select B5 by Osnap)  
 Specify second point of dis.. (Select B6 by Osnap)  
 Specify second point of dis.. (Select B7 by Osnap)  
 Specify second point of dis .. : (.j)

(10) Command: LINE (.j)

89 88-

Specify first point: (Select B8 by Osnap)  
 Specify next point or [Undo]: (Select B9 by  
 Osnap)  
 Specify next point or [Undo]: (.j)

(11) Command: LINE (.j)

Specify first point: (Select B10 by Osnap)  
 Specify next point or [Undo]: (Select B11 by  
 810--/ 811  
 Osnap)

Specify next point or [Undo]: (.j) FIG. 26--J 4(iv)

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(12) Command: LINE (.j)

Specify first point: (Select 812 by Osnap)  
 Specify next point or [Undo]: @70,7.5 (.j) (Point 813)  
 Specify next point or [Undo]: (Select 89 by Osnap)  
 Specify next point or [Close/Undo]: (.j)

(13) Command: LINE (.j)

Specify first point: (Select 815 by  
 Osnap) 70  
 Specify next point or [Undo]: @30,7.5  
 (.j) (Point 816)  
 Specify next point or [Undo]: (Select  
 814 by Osnap) 814  
 Specify next point or [Close/Undo]: 813  
 (.j)

(14) Command: -LAYER (.j)

Current layer: "O" 816  
 Enter an option 815  
 [?/Make/Set/New/ON/OFF/Color/Ltype/ FIG. 26-14(v)  
 LWeight ... J: SET (.j)  
 Enter layer name to make current <0>: TXT (.j)

Enter an option

[?/Make/Set/New/ON/OFF/Color/Ltype/LWeight ... ]: (.j)

(15) Command: DTEXT (.j)

Current text style: "Standard" Text height: 0.2000

Specify start point of text or [Justify/Style]: J (.j)

Enter an option [Align/Fit/Center/Middle/Right/TL/TC/TR/ML/MC/ ... ]: MC (.j)

Specify middle point of text: (Select M1 by Osnap)

Specify height <0.2000>: 2 (.j)

Specify rotation angle of text <0>: 0 (.j)

Enter text: NAME (.j)

Enter text: (.j)

(16) Command: DTEXT (.j)

Current text style: "Standard" Text height:

2.0000

Specify start point of text or [Justify/Style]: J

(.j)

Enter an option [Align/Fit/Center/Middle/Right/

TL/TC/TR/ML/MC/ ... ]: MC (.j) FIG. 26-14(vi)

Specify middle point of text: (Select M2 by Osnap)

Specify height <2.0000>: 3 (.j)

Specify rotation angle of text <0>: 0 (.j)

Enter text: CHAROTAR PUBLISHING HOUSE (.j)

Enter text: (.j)

(17) Command: DTEXT (.j)

Current text style: "Standard" Text height: 2.0000

Specify start point of text or [Justify/Style]: J (.j)

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Enter an option [Align/Fit/Center/Middle/Right/TL/TC/TR/MUMC/... ]: MC (.j)

Specify middle point of text: (Select M3 by Osnap)

Specify height <3.5000>: 3 (.j)

Specify rotation angle of text <0>: 0 (.j) tl.AME DATE

CHAROTAR PUBLISHING HOUSE

I

Enter text: SCALE (.j) ""

ll£N

THIS IS MY FIRST

TAP

Enter text: (.j) a;o AUTOCAD DRAWING

cc,,,

(18) Complete The Text Writing by DTEXT Command SCALE

(19) Command: -LAYER (.j)

Current layer: "0"

1:1 ORG\_f.0.:ML 0071:i.«!8

I

FIG. 26-14(vii)

Enter an option

[?/Make/Set/New/ON/OFF/Color/Ltype/LWeight/... ]: OFF (..)

Enter name list of layer(s) to turn off or < select objects>: GRID (..)

Enter an option

[?/Make/Set/New/ON/OFF/Color/Ltype/LWeight/ ... ]: (..)

(20) Save This File As Module 26-6.DWG

26-1

MODULE 26-6: USE OF LAYER, RECTANGLE, EXPLODE, TEXT AND COPY COMMANDS

30 70

10

t

LO NAME SIGN

ORG

CHAROTAR PUBLISHING HOUSE

OSN

THIS IS MY FIRST

CHO

TRO AUTOCAD DRAWING

LO APO

~L SCALE

1:1 ORG. NO. NIL 007/2009

26-I 4

on the diagram (fig. 26-15).

To generate Dimensions on the given diagram, follow the steps mentioned below. After executing the commands in sequence, we will get the output as shown in fig. 26-15(i) to fig. 26-15(iv).

(1) Open drawing file Module 26-5.DWG (fig. 26-13)

(2) Command: DIMSCALE (..)

Enter new value for DIMSCALE <1.0000>: 10 (..)

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(3) Command: DIMDEC (..)

Enter new value for DIMDEC <4>: 0 (..)

(4) Command: DIMEXO (..)

Enter new value for DIMEXO <).0625>: 0.2 (..)

(5) Command: DIMLINEAR (..)

Specify first extension line origin or <select object>: I I

s2] 84-J

(Select 81 by OSNAP)

FIG. 26-1 S(i)

Specify second extension line origin: (Select 82 by OSNAP)

Specify dimension line location or

[Mtext/Text/Angle/Horizontal/Vertical/Rotated]: (Select Point

Below 81 or 82)

Dimension text = 25

(6) Command: DIMCONTINUE (..) )

, 25

Specify a second extension line origin or [Undo/Select] , -+---->+-<---40 ' 25 )>--j,

<Select>: (Select 83 by OSNAP)

FIG. 26-1 S(ii)

Dimension text = 40

Specify a second extension line origin or [Undo/Select] <Select>: (Select 84 by

OSNAP)

Dimension text = 25

Specify a second extension line origin or [Undo/Select] <Select>: (..)

Select continued dimension: (..)

(7) Command: DIMLINEAR (..)

Specify first extension line origin or <select object>: (Select B4 by OSNAP)

Specify second extension line origin: (Select 85 by OSNAP)

Specify dimension line location or

[Mtext/Text/Angle/Horizontal/Vertical/Rotated]: (Click on Right of 84)

Dimension text = 20

(8) Command: DIMBASELINE (..)

Specify a second extension line origin or [Undo/

Select] <Select>: (Select 86 by OSNAP)

Dimension text = 35 t

Specify a second extension line origin or [Undo/

Select] <Select>: (..)

Select base dimension: (..) 84

(9) Command: DIMDIAMETER (..) Fig. 26-1 S(iii)

Select arc or circle: (Select Circle C1)

Dimension text = 40

Specify dimension line location or [Mtext/Text/Angle]: (Click Outside C1)

(10) Command: DIMDIAMETER (..)

Select arc or circle: (Select Circle C2)

Dimension text = 20

Specify dimension line location or [Mtext/Text/Angle]: (Click Outside C2)

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(11) Command: DIMRADIUS (.J)

Select arc or circle: (Select Arc A1)

Dimension text = 5

Specify dimension line location or [Mtext/Text/

Angle]: (Click Outside A1)

(12) Similarly complete the other dimensions.

(13) Save This File As Module 26-7.DWG

2 B4

of 26• 7 (fig. 2 6-1 5): ~---s-i)Ml....( - - - - , . - , . + - - -

MODULE 26-7: USE OF DIMENSIONS FIG. 26-1 S(iv)

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FIG. 26-15

Module 26-8. Draw a line as shown in fig. 26-16. Use Polygon 1 Chamfe(

Arc and Ivfirror

To draw a line diagram using Polygon, Chamfer, Arc and Mirror commands, follow the steps mentioned below. After executing the commands in sequence, we will get the output as shown in fig. 26-16(i) to fig. 26-16(v).

(1) Command: LIMITS (.J)

Reset Model space limits:

Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (.J)

Specify upper right corner <12.0000,9.0000>: 120,90 (.J)

(2) Command: ZOOM (.J)

Specify corner of window, enter a scale factor (nX or nXP), or

[All/Center/Dynamic/Extents/Previous/Scale/Window] < real time>: ALL (.J)

Regenerating model.

(3) Command: LINE (.J) (30,65)

P0 1 ~

Specify first point: 30,65 (.J) ~

Specify next point or [Undo]: 20 (.J) (When 0° POLAR is ON) FIG. 26-1 6(i)

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Specify next point or [Undo]: 10 (.J) (When 270° POLAR is ON)

Specify next point or [Close/Undo]: 20 (.J) (When 0° POLAR is ON)

Specify next point or [Close/Undo]: 10 (.J) (When 90° POLAR is ON)

Specify next point or [Close/Undo]: 20 (.J) (When 0° POLAR is ON)

Specify next point or [Close/Undo]: (.J)

(4) Command: ARC (.J)

Specify start point of arc or [Center]: (Click Point P1 When ENDPOINT magnet gets ON)

Specify second point of arc or [Center/End]: E (.J)

Specify end point of arc: @0,-40 (.J) (Point P2) L3

Specify center point of arc or [Angle/Direction/Radius]:

A (.J)

Specify included angle: 180 (.J)

(5) Command: CHAMFER (.J)

(TRIM mode) Current chamfer Dist1 = 0.5000, Dist2

= 0.5000 P2

Select first line or [Polyline/Distance/Angle/Trim/Method]:

D (.J) FIG. 26-16(ii)

Specify first chamfer distance <0.5000>: 4 (.J)

Specify second chamfer distance <4.0000>: 4 (.J)

Select first line or [Polyline/Distance/Angle/Trim/Method]: (Select Line L2)

Select second line: (Select Line L3)

(6) Command: CHAMFER (.J)

(TRIM mode) Current chamfer Dist1 = 4.0000, Dist2 = 4.0000

Select first line or [Polyline/Distance/Angle/Trim/Method]: (Select Line L3)

Select second line: (Select Line L4)

(7) Command: MIRROR (.J)

Select objects: (Select Arc A1)

Select objects: (.J)

Specify first point of mirror line: (Click MIDPOINT magnet of Line L3)

Specify second point of mirror line: (Click When

270° POLAR is ON) FIG. 26-16(iii)

Delete source objects? [Yes/No] <N>: N (.J)

(8) Command: MIRROR (.J)

Select objects: (Select All Objects in Dotted BOX)

Select objects: (.J)

Specify first point of mirror line: (Click MIDPOINT of Arc A1)

Specify second point of mirror line: (Click

MIDPOINT of Arc A2)

Delete source objects? [Yes/No] <N>: N (.J)

FIG. 26-16(iv)

(9) Command: POLYGON (.J)

Enter number of sides <4 >: 6 (.J)

Specify center of polygon or [Edge]: (Select CENTER I CENTER Magnet magnet of Arc A1) ...--~-,---, L3

Enter an option [Inscribed in circle/Circumscribed about circle] <I>: I (.J)

Specify radius of circle: 8 (.J)

(10) Command: MIRROR (.J)

H1 0

Select objects: (Select Hexagon H1)

Select objects: (.J) FIG. 26-16(v)

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Specify first point of mirror line: (Click MIDPOINT of L3)

Specify second point of mirror line: (Click When 270° POLAR is ON)

Delete source objects? [Yes/No] <N>: N (.J)

(11) Command: ERASE (.J)

Select objects: ALL (.J)

Select objects: (.J)

(12) Command: U (.J)

(13) Save This File As Module 26-8.DWG

Output of Module 26-8 (fig. 26-16):

MODULE 26-8: USE OF POLYGON, CHAMFER, ARC AND MIRROR COMMANDS

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I I

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12

> I 41 <

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FIG. 26-16

Module 26-9. Draw a line diagram as shown in fig. 26-'17. Use Polyline command.

To draw a line diagram using Polyline command, follow the steps mentioned below.

(1) Command: LIMITS (.J)

Reset Model space limits:

Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (.J)

Specify upper right corner <12.0000,9.0000>: 120,90 (.J)

(2) Command: ZOOM (.J)

Specify corner of window, enter a scale factor (nX or nXP), or

[All/Center/Dynamic/Extents/Previous/Scale/

Window] <real time>: ALL (.J)

Regenerating model.

(3) Command: PLINE (.J)

Specify start point: 15,20 (.J)

Current line-width is 0.0000

Specify next point or [Arc/Halfwidth/Length/

Undo/Width]: WIDTH (.J) P1(15,20)

Specify starting width <0.0000>: 0 (.J)

FIG. 26-17(i)

Specify ending width <0.0000>: 10 (.J)

Specify next point or [Arc/Halfwidth/Length/ ... ]: 30 (.J) (When 0° POLAR is ON)

Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]: WIDTH (.J)

Specify starting width <10.0000>: 2.5 (.J)

Specify ending width <2.5000>: 2.5 (.J)

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Specify next point or [Arc/Close/Halfwidth/... ]: 50 (.J) (When 0° POLAR is ON)

Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]: ARC (.J)

Specify endpoint of arc or

[Angle/CEnter/CLose/Direction/Halfwidth/... ]: 40 (.J) (When 90° POLAR is ON)

Specify endpoint of arc or

[Angle/CEnter/CLose/Direction/Halfwidth/Line/Radius/ ... ]: LINE (.J)

Specify next point or [Arc/Close/Halfwidth/ ... ]: 50 (.J) (When 180° POLAR is ON)

Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]: WIDTH (.J)

Specify starting width <2.5000>: 10 (.J)

Specify ending width <10.0000>: 0 (.J)

Specify next point or [Arc/Close/Halfwidth/ ... ]: 30 (.J) (When 180° POLAR is ON)

Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]: (.J)

(4) Save this File As Module 26-9.DWG

of Module 26-9 (fig. 26-17):

MODULE 26-9: USE OF POLYLINE COMMAND

10WIDTH 2.5WIDTH

R20

I. 50

FIG. 26-1 7

Module 26-1 O. Draw a line diagram as shown in 26-JB. Use Donut and Rectangular Array, commands.

To draw a line diagram using Donut and Rectangular Array commands, follow the steps mentioned below. After executing the commands in sequence, we will get the output as shown in fig. 26-18(i) to fig. 26-18(iii).

(1) Command: LIMITS (.J)

Reset Model space limits:

Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (.J)

Specify upper right corner <12.0000,9.0000>: 120,90 (.J)

(2) Command: ZOOM (.J)

Specify corner of window, enter a scale factor (nX or nXP), or

[All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>: ALL (.J)

(3)

Regenerating model.

Command: RECTANGLE (.J)

Specify first corner point or [Chamfer/Elevation/Fillet/

Thickness/Width]: 10,20 (.J) (

J

~1o)

100

@100,

501 t50t

1

Specify other corner point or [Dimensions]: -----~

@100,50 (.J) FIG. 26-1

Art. 26-7] Computer-Aided Drafting 651

(4) Command: DONUT (.J)

Specify inside diameter of donut <0.5000>: 4 (.J)

:~fDDNUT

Specify outside diameter of donut <1.0000>: 6 (.J)

Specify center of donut or <exit>: FROM (.J) t L.c-10- - - - '

-)--j--j-,t-

Base point: (Select R1 Point by ENDPOINT magnet) <Offset>:

FIG. 26-18(ii)

@10,10 (.J)

0 0 0 0 0

Specify center of donut or <exit>: (.J)

LO

0 0 0 0 0

(5) Command: ARRAY (.J)

0 0 0 0

Select objects: (Select Donut D1) 1

D1

Select objects: (.J)

(1) Select Rectangulaar area

(2) Click here and select object

i.e. donut

(3) Enter No. of columns= 5

(4) Enter No. of rows= 3

(5) Enter row distance= 15

(6) Enter column distance= 20

(7) Finally click OK

FIG. 26-18(iv)

(6) Save this File As Module 26-10.DWG

Output of Module 26-10 (fig. 26-18):

MODULE 26-10: USE OF DONUT AND RECTANGULAR ARRAY COMMANDS

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20 20

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ROW3 -G-

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n:: u.

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I

ROW 2 -t--1--H

11-

-G-

I

~II~

n:: u.

10

ROW 1 -'---1--->-1

ICOLUMN ! COLUMN I COLUMN

r--

OFFSETT- OFFSET ~ ~ OFFSET

COLUMN 1 COLUMN 2 COLUMN 3

1

COLUMN

OFFSET

COLUMN 4 COLUMN 5

FIG. 26-18

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26-11. Draw a diagram as shown in 26-19. Use and Polar Array commands.

To draw a diagram using Ellipse and Polar Array commands, follow the steps mentioned below. After executing the commands in sequence, we will get the output as shown in fig. 26-19(i) to fig. 26-19(iii).

(1) Command: LIMITS (.J)

Reset Model space limits:

Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (.J)

Specify upper right corner <12.0000,9.0000>: 120,90 (.J)

(2) Command: ZOOM (.J)

Specify corner of window, enter a scale factor (nX or nXP), or

[All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>: ALL (.J)

Regenerating model. 080

(3) Command: CIRCLE (.J)

Specify center point for circle or [3P/2P/Ttr (tan tan radius)]: 60,45 (.J)

Specify radius of circle or [Diameter]: D (.J) \

Specify diameter of circle: 80 (.J) C1 \

(4) Command: CIRCLE (.J) -{Z60,45) )

Specify center point for circle or [3P/2P/Ttr (tan tan radius)]: (Select CENTER magnet C1 of 080 \ I Circle) /

Specify radius of circle or [Diameter] <40.0000>: ./

D (.J)

Specify diameter of circle <80.0000>: 60 (.J)

(5) Command: ELLIPSE (.J)

Specify axis endpoint of ellipse or [Arc/Center]: C (.J)

Specify center of ellipse: (Select 90° QUADRANT Q1 magnet of 060 Circle)

Specify endpoint of axis: 6 (.J) (when 90° POLAR is I

~c9\

ON)

(6)

Specify distance to other axis or [Rotation]: 3 (.J)

Command: ARRAY (.J)

4 c1 (p

Specify center point of array: (Select CENTER magnet C1)

Select objects: (Select Ellipse E1)

Select objects: (.J)

FI( . ? r. .1 9

(1) Click Polar Array ' - ., -

(2) Click here and select object as Ellipse

(3) Click here and select Center C1

(4) Enter .\Jo. of Items i.e. 8

(5) Finally click OK

Art. 26-7] Computer-Aided Drafting 653

(7) Command: DIMSCALE (.J) CR1

Enter new value for DIMSCALE <1.0000>: 10 (.J)

(8) Command: DIMCENTER (.J)

Select arc or circle:(Select Circle CR1)

(9) Command: EXTEND

Current settings: Projection=UCS, Edge=None

Select boundary edges ...

Select objects or < select all>: (Select Circle CR1 )

Select objects: (.J)

Select object to extend or shift-select to trim or

[Fence/Crossing/Project/Edge/Undo]:(Click Right End of Line L1)

Select object to extend or shift-select to trim or FIG. 26-19(iv)

[Fence/Crossing/Project/Edge/Undo]: (Click Left End of Line L1)

Select object to extend or shift-select to trim or

[Fence/Crossing/Project/Edge/Undo]: (Click Top End of o< \ Line L2)

Select object to extend or shift-select to trim or

[Fence/Crossing/Project/Edge/Undo]: (Click Bottom End of Line L2)

Select object to extend or shift-select to trim or

[Fence/Crossing/Project/Edge/Undo]: (.J)

oz /

(10) Save This File As Module 26-11.DWG

FIG. 26-19(v)

Output of Module 26-11 (fig. 26-19):

## MODULE 26-11: USE OF ELLIPSE AND POLAR ARRAY COMMANDS

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I \

CD \

CD I

FIG. 26-19

Module 26-12. Draw a line diagram as shown in fig. 26-20. Use of Hatch, Rotate, Stretch and Scale commands.

To draw a line diagram using Hatch, Rotate, Stretch and Scale commands, follow the steps mentioned below. After executing the commands in sequence, we will get the output as shown in fig. 26-20(i) to fig. 26-20(vi).

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(1) Command: LIMITS (.J)

Reset Model space limits:

Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (.J)

Specify upper right corner <12.0000,9.0000>: 120,90 (.J)

(2) Command: ZOOM (.J)

Specify corner of window, enter a scale factor (nX or nXP), or

[All/Center/Dynamic/Extents/Previous/Scale/Window] <real time>: All (.J)

(3) Command: POLYGON (.J)

Enter number of sides <4>: 5 (.J)

Specify center of polygon or [Edge]: E (.J)

Specify first endpoint of edge: 45,40 (.J)

Specify second endpoint of edge: 30 (.J) (When 0°

POLAR is ON) 30

f+---~i

(4) Command: BHATCH (.J)

FIG.

(3) Click here to select  
internal point

(1) Click here and select  
ANSI31 pattern

(2) Enter scale= 10

(4) Finally click OK

FIG.

Select internal point: (Click Inside Pentagon)

Selecting everything visible .. .

Analyzing the selected data .. .

Analyzing internal islands ...

Select internal point: (.J)

< Hit enter or right-click to return to the dialog> (.J)

(5) Command: ROTATE (.J)

Current positive angle in UCS: ANGDIR = counterclockwise

ANGBASE=0

Select objects: (Select Pentagon)

Select objects: (Select Hatch Pattern)

Select objects: (.J)

Specify base point: (Select MIDPOINT magnet M1)

Specify rotation angle or [Reference]: - 90 (.J) FIG.

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(6) Command: STRETCH (..J)

Select objects to stretch by crossing-window or crossing-polygon ...

Select objects: (Click near to S1 and then click near to S2) M2

Select objects: (.J)

Specify base point or displacement: (Select ENDPOINT magnet 82)

Specify second point of displacement or < ... >:

40 (.J) (When 0° POLAR is ON)

(7) Command: SCALE (.J) FIG. 26-20(v)

Select objects: (Select Pentagon)

Select objects: (Select Hatch Pattern) M2~

Select objects: (.J)

Specify base point: (Select MIDPOINT magnet M2)

~|~

43

Specify scale factor or [Reference]: 0.5 (.J)

(8) Save This File As Module 26-12.DWG FIG. 26-20(vi)

of Module 26-12 (fig.

MODULE 26-12: USE OF HATCH, ROTATE, STRETCH AND SCALE COMMANDS

FIG. 26-20

Module 26-13. Draw a 2D diagram as shown in fig. 26-21.

To draw a 2D diagram using the steps mentioned below. After executing the commands in sequence, we will get the output as shown in fig. 26-21 (i) to fig. 26-21 (x).

(1) Command: LIMITS (.J)

Reset Model space limits:

Specify lower left corner or [ON/OFF] <0.0000,0.0000>: (.J)

Specify upper right corner <12.0000,9.0000>: 180,135 (.J)

(2) Command: ZOOM (.J)

Specify corner of window, enter a scale factor (nX or nXP), or

[All/Center/Dynamic/Extents/Previous/Scale/ .... ] <real time>: ALL (.J)

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(3) Command: LTSCALE (.J)

Enter new linetype scale factor <1.0000>: 15 (.J)

(4) Command: -LAYER (.J)

Enter an option

[?/Make/Set/New/ON/OFF/Color/Ltype/LWeight/Plot/Freeze/... J]: NEW (.J)

Enter name list for new layer(s): CENLIN (.J)

Enter an option

[ ?/Make/Set/New/ON/OFF/Color/Ltype/LWeight/Plot/Freeze/... ]: COLOR (.J)

New color [Truecolor/Colorbook] : 41 (.J)

Enter name list of layer(s) for color 41 <O>: CENLIN (.J)

Enter an option

[ ?/Make/Set/New/ON/OFF/Color/Ltype/LWeight/Plot/Freeze/... ] : LTYPE (.J)

Enter loaded linetype name or [?] <Continuous>: DASHDOT (.J)

Enter name list of layer(s) for linetype "DASHDOT" <O>: CENLIN (.J)

Enter an option

[?/Make/Set/New/ON/OFF/Color/Ltype/LWeight/Plot/Freeze/... J]: (.J)

(5) Command: CIRCLE (.J)

Specify center point for circle or [3P/2P/Ttr (tan tan radius)]:

90,100 (.J)

Specify radius of circle or [Diameter]: d (.J) ~90,100

Specify diameter of circle: 68 (.J)

(6) Command: CIRCLE (.J)

Specify center point for circle or [3P/2P/Ttr (tan tan radius)]:

90,100 (.J) FIG. 26-21 (i)

Specify radius of circle or [Diameter] <3.4000>: d (.J)

Specify diameter of circle <6.8000>: 26 (.J)

(7) Command: LINE (.J)

Specify first point: (Select QUADRANT Magnet Q2 of Circle 068)

Specify next point or [Undo]: (Type 121 and press enter when 270°

Specify next point or [Undo]: (Type 40 and press enter when 0°

Polar is ON)

Specify next point or [Close/Undo]: (Type 34 and press enter when 90° Polar is ON)

Specify next point or [Close/Undo]: (Type 80 and press enter

when 180° Polar is ON)

80

Specify next point or [Close/Undo]: (Type 34 and press enter when 270° Polar is ON)

Specify next point or [Close/Undo]: (Select ENDpoint Magnet E1)

Specify next point or [Close/Undo]: (.J)

(8) Command: FILLET (.J) FIG. 26-21 (ii)

Select first object or [Undo/Polyline/Radius/ ... J]:(Select Line L1 from Right end)

Select second object : (Select Line L2 from Right end)

(9) Command: FILLET (.J)

Select first object or [Undo/Polyline/Radius/... J]:(Select Line L1 from Left end)

Select second object : (Select Line L2 from Left end)

(10) Command: CIRCLE (.J)

Specify center point for circle or [3P/2P/ ... J]: (Select MIDpoint Magnet M1 of Line L3) FIG. 26-21 (iii)

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Specify radius of circle or [Diameter] <1.7000>: d (.J)

Specify diameter of circle <3.4000>: 16 (.J)

(11) Command: CIRCLE (.J)

Specify center point for circle or [3P/2P/ ... J]: (Select MIDpoint Magnet M2 of Line L4) L3

Specify radius of circle or [Diameter] < 1. 7000>: d M 2 M1

(.J)

Specify diameter of circle <3.4000>: 16 (.J)

(12) Command: LINE (.J) FIG. 26-21 (iv)

Specify first point: (Select MIDpoint Magnet M3 of Arc A1)

Specify next point or [Undo]: (Type 34 and press enter when 180° Polar is ON)

Specify next point or [Undo]: (.J)

(13) Command: LINE (.J)

Specify first point: (Select MIDpoint Magnet M4 of Arc A2) A

2 A1

Specify next point or [Undo]: (Type 34 and press enter when 0° Polar is ON)

Specify next point or [Undo]: (.J)

(14) Command: OFFSET (.J)

Specify offset distance or [Through/Erase/Layer] FIG. 26-21 (v)

<Through>: 15 (.J)

Select object to offset or [Exit/Undo] <Exit>: (Select Line L5)

Specify point on side to offset or [Exit/ .. ] <Exit>: (Click on Right Side of Line L5)

Select object to offset or [Exit/Undo] <Exit>: (Select Line L5)

Specify point on side to offset or [Exit/.. ] <Exit>: (Click on Left Side of Line L5)

Select object to offset or [Exit/Undo] <Exit>: (.J)

(15) Command: FILLET (.J)

Select first object or [Undo/Polyline/Radius/Trim/Multiple]: r (.J)

Specify fillet radius <0.0000>: 10 (.J)

FIG. 26-21 (vi)

Select first object or [Undo/Polyline/Radius/... ] : (Select Line L6)

Select second object or shift-select to apply corner:

(select Circle CR1)

(16) Command: FILLET (.J)

Select first object or [Undo/Polyline/Radius/Trim/Multiple]: r (.J)

Specify fillet radius <0.0000>: 10 (.J)

Select first object or [Undo/Polyline/Radius/ ... ] : (Select CR1 Line L7)

Select second object or shift-select to apply corner:

(Select Circle CR1) L7 L6

(17) Command: TRIM (.J) L1

Current settings: Projection=UCS, Edge=Extend

Select cutting edges ...

Select objects or <select all>: (Select Line L1)

Select objects: (.J) FIG. 26-21(vii)

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Select object to trim or shift-select to extend or

[Fence/ ..... /Undo]: (Select Line L6 from below Line L1)

Select object to trim or shift-select to extend or

[Fence/..... /Undo]: (Select Line L7 from below Line L1)

Select object to trim or shift-select to extend or

[Fence/Crossing/Project/Edge/eRase/Undo]: (.J)

(18) Command: TRIM (.J)

Current settings: Projection= UCS, Edge= Extend FIG. 26-21 (viii)

Select cutting edges ...

Select objects or < select all>: (Select Line L6)

Select objects: (Select Line L7)

Select objects: (..J)

Select object to trim or shift-select to extend or L6

[Fence/.. ]: (Select Line L1 from between Line L6 and L7)

Select object to trim or shift-select to extend or

[Fence/Crossing/Project/Edge/eRase/Undo]: (..J)

(19) Command: LINE (..J) FIG. 26-2"1 (ix)

Specify first point: (Select QUADRANT Magnet Q1 of Circle 068)

Specify next point or [Undo]: (Select QUADRANT Magnet Q3 of Circle 068)

Specify next point or [Undo]: (..J)

(20) Select Line L3,L4,L5,L8, L9 and L10 by Mouse and Click

in Layer tool box and

select CENLIN Layer and Press Esc key.

FIG. 26-21 (x)

(21) Save This Fi le As Module 26-13.DWG Module 26-14. Draw 20 diagram

of Module 26-13 26-2·1 ): as shown in fig. 26-22(ii) and

26-22(iii). Using BLOCK and INSERT

068

command create assembly as shown in

fig. 26-22(vii)

(1) Use LINE command to draw the

two given drawings as shown in

fig. 26-22(ii) and fig. 26-22(iii).

Use start point (1, 1) and (8, 1) for

0 each respectively.

R17 (2) Command: BLOCK (..J)

(1) The Dialog Box Of BLOCK will

Open as shown in fig. 26-22(i).

Enter the name 'B' in NAME

box. Type 8.4 in X box and 2

in Y box. Select Delete radio

button and finally click Select

FIG. 26-21 Objects button.

Art. 26-8] 659

FIG. 26-22(i)

(2) Select objects: (Select All Objects of 1.8

PIN) Fig. 26-22(iii) 1:7 "'- ~--+j J.

Select objects: (+J)

(3) Finally click OK button. =lk(1,1) ~ (8,1)-U::"t

(3) Command: OOPS (+J) ~.~~-6-----J ~~

(4) Command: INSERT (+J) BODY PIN

(1) The Dialog Box Of INSERT will Open FIG. 26-22(ii) Fie. 26-22 as shown below in fig. 26-22(iv).

(2) Select 'B' block from NAME box. Enter 1 in X box. Enter O in Rotation box. Finally click OK button.

j"1t

(8.4,1)

FIG. 26-22(iv) FIG. 26-22 FIG. 26-22

(3) Specify insertion point or [Basepoint/Scale/Rotate]: (Select MIDpoint M1)

(5) Similarly Insert Block 'B' at MIDpoint M2 and M3 with X value of 0.5 and Rotation angle of -45° and 45° respectively.

FIG. 26-22

(6) Save this drawing file as Module 26-14.DWG

(i) Isometric drawings are used to help visualize the shape of an object. It is just a two dimensional representation of a 3D drawing in a 2D plane.

(ii) An isometric view is obtained by rotating the object 45° around the imaginary vertical axis and then tilting the object upward through a 35 degree and 16 minute angle.

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(iii) As shown in fig. 26-23, the three principal axes are AB (30° inclined with horizontal), AC (30° inclined with horizontal) and AD (90° inclined with horizontal). They also make 120° angle with each other.

(iv) Isometric drawings have three principal planes, isoplane right, isoplane top and isoplane left as shown in fig. 26-23.

(1) set 26-15, 26-16)

(i) Use the snap command to set the isometric grid and snap. The isometric grid lines are displayed at 30° of the horizontal axis and also set the vertical spacing between grid.

(ii) The command sequence is as below: A

Command: -GRID (.J) FIG. 26-23

Specify grid spacing or [ON/OFF/ ... ] <1.0000>: 1 (.J)

Command: -GRID (.J)

Specify grid spacing or [ON/OFF/ ... ] <1.00000>: ON (.J)

Command: -SNAP (.J)

Specify snap spacing or [ON/OFF/ ... /style ... ] <0.5000>: 5 (.J)

Enter snap grid style [Standard/Isometric] <S>: I (.J)

Specify vertical spacing <0.5000>: 1 (.J)

(iii) The cross hair now will be displayed at an isometric angle.

set 26-1 26-1

(i) There are three orientation of crosshairs, isotop, isoright and isoleft.

(ii) These orientation are set by isoplane command, the sequence is as below:

Command: ISOPLANE (.J)

Enter isometric plane setting [Left/Top/Right] <Top>: T (.J)

Current Isoplane: Top

(iii) We can also toggle among different isoplanes using the f5 function key.

module 26-1

(i) Isometric circles are drawn in three planes i.e. isotop, isoright and isoleft.

(ii) So first set the isoplane using ISOPLANE command.

(iii) Now use isocircle option of ELLIPSE command to create isometric circle. The sequence is as below:

Command: ELLIPSE (.J)

Specify axis endpoint of ellipse or [Arc/Center/Isocircle]: I (.J)

Specify centre of isocircle: (Select a point)

Specify radius of isocircle or [Diameter]: (Enter radius value)

26-1

(i) Isometric text are created in three isoplanes.

(ii) These text are obliqued at 30° or 330° angle with rotation angle of 30° or 330°.

(iii) The oblique angle and rotation angle for isotext in three planes are given below:

Isoplane Oblique angle Rotation angle

Top 30 330

Right 30 30

Left 330 330

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(iv) For this use STYLE command to create two text styles (iso 30 and iso 330) having 30° and 330° oblique angle as shown in module 26-18.

(v) Set the isoplane and use DT (Dynamic Text) command to write text in respective plane.

(i) Dimension the distance using DIMALIGNED command.

(ii) Change the oblique angle of dimension by using oblique option of DIMEDIT command. Note: The oblique angle is determined by the angle the extension line of the isometric dimension makes with the positive X-axis.

26-15. Draw a isometric

and

(1) Command: GRID (.J)

Specify grid spacing(X) or [ON/OFF/Snap/Major/aDaptive/... ] <0.5000>: 1 (.J)

(2) Command: GRID (..) )

Specify grid spacing(X) or [ON/OFF/Snap/Major/aDaptive/ ... ] < 1.0000>: ON (..) )

(3) Command: SNAP (..) )

Specify snap spacing or [ON/OFF/Aspect/Style/Type] <0.5000>: STYLE (..) )

Enter snap grid style [Standard/Isometric] <S>: ISOMETRIC (..) )

Specify vertical spacing <0.5000>: (..) )

(4) Command: SNAP (..) )

Specify snap spacing or [ON/OFF/Style/Type] <0.5000>: OFF (..) )

(5) Command: ISOPLANE (..) )

Current isoplane: Right

Enter isometric plane setting [Left/Top/Right] <Left>: RIGHT (..) )

Current isoplane: Right

(6) Command: LINE (..) )

Specify first point: 6, 1 (..) )

Specify next point or [Undo]: 4 (..) ) (When 30° POLAR is ON)

Specify next point or [Undo]: 4 (..) ) (When 90° POLAR is ON)

Specify next point or [Close/Undo]: 4 (..) ) (When 210° POLAR is ON)

Specify next point or [Close/Undo]: C (..) ) (6, 1) 30°

(7) Command: ISOPLANE (..) ) ~

Current isoplane: Right FiG.

Enter isometric plane setting [Left/Top/Right] <Left>:

LEFT (..) )

Current isoplane: Left

(8) Command: LINE (..) )

Specify first point: 6, 1 (..) )

Specify next point or [Undo]: 4 (..) ) (When 150° POLAR is ON)

Specify next point or [Undo]: 4 (..) ) (When 90° POLAR is ON)

Specify next point or [Close/Undo]: 4 (..) ) (When 330° POLAR is ON)

Specify next point or [Close/Undo]: (..) )

FIG.

(9) Command: ISOPLANE (..) )

Current isoplane: Left

Enter isometric plane setting [Left/Top/Right] <Top>: TOP (..) )

Current isoplane: Top

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(10) Command: LINE (.J)

Specify first point: (Select ENDPOINT magnet E1)

Specify next point or [Undo]: 4 (.J) (When 150°

POLAR is ON)

Specify next point or [Undo]: 4 (.J) (When 210°

POLAR is ON)

Specify next point or [Close/Undo]: (.J)

(11) Command: SNAP (.J)

Specify snap spacing or [ON/OFF/Aspect/Style/Type]

<0.5000>: STYLE (.J)

Enter snap grid style [Standard/Isometric] <S>:

STANDARD (.J)

Specify vertical spacing <0.5000>: (.J) FIG. 26-24(iii)

(12) Command: SNAP (.J)

Specify snap spacing or [ON/OFF/Style/Type] <0.5000>: OFF (.J)

(13) Save this File As Module 26-15.DWG

Module 26-16. Open the file Module 26-15.DWG and create fsocircles in each isoplane using Ellipse command.

(1) Open The File Module 26-15.DWG

(2) Command: GRID (.J)

Specify grid spacing(X) or [ON/OFF/Snap/Major/aDaptive/... ] <0.5000>: 1 (.J)

(3) Command: GRID (.J)

Specify grid spacing(X) or [ON/OFF/Snap/Major/aDaptive/ ... ] <1.0000>: ON (.J)

(4) Command: SNAP (.J)

Specify snap spacing or [ON/OFF/Aspect/Style/Type] <0.5000>: STYLE (.J)

Enter snap grid style [Standard/Isometric] < S>: ISOMETRIC (.J)

Specify vertical spacing <0.5000>: (.J)

(5) Command: SNAP (.J)

Specify snap spacing or [ON/OFF/Style/Type] <0.5000>:

OFF (.J)

(6) Command: LINE (.J)

Specify first point: (Select ENDPOINT magnet E1)

Specify next point or [Close/Undo]: (Select ENDPOINT magnet E2)

Specify next point or [Close/Undo]: (Select ENDPOINT magnet E3)

Specify next point or [Close/Undo]: (.J)

(7) Command: ISOPLANE (.J)

Current isoplane: Left

Enter isometric plane setting [Left/Top/Right] <Top>:

TOP (.J)

Current isoplane: Top

(8) Command: ELLIPSE (.J)

Specify axis endpoint of ellipse or [Arc/Center/Isocircle]:

I (.J)

Specify center of isocircle: (Select MIDPOINT magnet M1 by mouse)

Specify radius of isocircle or [Diameter]: 1.25 (.J)

(9) Command: ISOPLANE (.J)

Current isoplane: Top

Enter isometric plane setting [Left/Top/Right] <Right>:

RIGHT (.J)

Current isoplane: Right FIG. 26-25(ii)

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(10) Command: ELLIPSE (.J)

Specify axis endpoint of ellipse or [Arc/Center/ socircle]: I (.J)

Specify center of isocircle: (Select MIDPOINT magnet M2 by mouse)

Specify radius of isocircle or [Diameter]: 1.25 (.J)

(11) Command: ISOPLANE (.J)

Current isoplane: Right

Enter isometric plane setting [Left/Top/Right] <Left>: LEFT (.J)

Current isoplane: Left

(12) Command: ELLIPSE (.J)

Specify axis endpoint of ellipse or [Arc/Center/ socircle]: I (.J)

Specify center of isocircle: (Select MIDPOINT magnet M3 by mouse)

Specify radius of isocircle or [Diameter]: 1.25 (.J)

(13) Command: SNAP (.J)

Specify snap spacing or [ON/OFF/Aspect/Style/Type] <0.5000>: STYLE (.J)

Enter snap grid style [Standard/ sometric] <S>: STANDARD (.J)

Specify vertical spacing <0.5000>: (.J)

(14) Command: SNAP (.J)

Specify snap spacing or [ON/OFF/Style/Type] <0.5000>: Off (.J)

(15) Save this File As Module 26-16.DWG

26-17. Open the file Module 26-15.0WG and create Aligned dimensions.

(1) Open the file Module 26-15.DWG

(2) Command: DIMDEC (.J)

Enter new value for DIMDEC <4>: 1 (.J)

(3) Command: DIMALIGNED (.J)

Specify first extension line origin or <select object>: (Select ENDpoint E1)

Specify second extension line origin: (Select ENDpoint E2)

Specify dimension line location or

[Mtext/Text/Angle]: (Click below line L1)

Dimension text = 4.0000

FIG. 26-26(i)

(4) Command: DIMEDIT (.J)

Enter type of dimension editing [Home/New/Rotate/Oblique]

<Home>: 0 (.J)

Select objects: (Select dimension D1)

Select objects: (.J)

Enter obliquing angle (press ENTER for none): 90 (.J)

(5) Command: DIMALIGNED (.J)

Specify first extension line origin or <select object>: (Select  
ENDpoint E3)

Specify second extension line origin: (Select ENDpoint E4)

Specify dimension line location or

[Mtext/Text/Angle]: (Click Right of line L2) FIG. 26-26(ii)

Dimension text = 4.0000

(6) Command: DIMEDIT (.J)

Enter type of dimension editing [Home/New/Rotate/Oblique]

<Home>: 0 (.J)

Select objects: (Select dimension D2)

Select objects: (.J)

Enter obliquing angle (press ENTER for none): 30 (.J)

(7) Command: DIMALIGNED (.J)

Specify first extension line origin or <select object>: (Select  
E5)

Specify second extension line origin: (Select ENDpoint E6) FIG. 26-26(iii)

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Specify dimension line location or

[Mtext/Text/Angle]: (Click Left of line L3)

Dimension text = 4.0000

(8) Enter type of dimension editing [Home/New/Rotate/Oblique]

<Home>: 0 (+J)

Select objects: (Select dimension D3)

Select objects: (+J)

Enter obliquing angle (press ENTER for none): 150 (+J)

(9) Save this File As Module 26-17.DWG

FIG. 26-26(iv)

FIG. 26-26(v) FIG. 26-26(vi)

Module 26-18. Open the file Module 26-1 S.DWG and create Isotext in each  
plane as shown in fig. 26-27(iii).

(1) Open the File Module 26-15.DWG

(2) Command: SNAP (+J)

Specify snap spacing or [ON/OFF/Style/Type] <0.5000>: S (+J)

Enter snap grid style [Standard/Isometric] <I>: I (+J)

Specify vertical spacing <0.5000>: (+J)

(3) Command: SNAP (+J)

Specify snap spacing or [ON/OFF/Style/Type] <0.5000>: OFF (+J)

(4) Command: ISOPLANE (+J)

Current isoplane: Left

Enter isometric plane setting [Left/Top/Right] <Top>: TOP (+J)

Current isoplane: Top

(5) Command: -STYLE (+J)

Enter name of text style or [?] <IS0330>: IS030 (+J)

Specify full font name or font filename (TIF or SHX) <ROMANS.shx>: ROMANTIC  
(+J)

Specify height of text or [Annotative] <0.2000>: 0.18 (+J)

Specify width factor < 1.0000>: (+J)

Specify obliquing angle <0>: 30 (+J)

Display text backwards? [Yes/No] <No>: (+J)

Display text upside-down? [Yes/No] <No>: (+J)

Vertical? [Yes/No] <No>: (+J)

"IS030" is now the current text style.

(6) Command: DT (+J)

Current text style: "IS030" Text height: 0.1800 Annotative:  
No

Specify start point of text or [Justify/Style]: J (+J)

Enter an option [Align/Fit/Center/Middle/Right/TL/TC/TR/MU  
MC/MR/BL/BC/BR]: BC (+J)

Specify bottom-center point of text: (Select MIDpoint M1)

Specify rotation angle of text <0>: 330 (+J)

Enter text: CHAROTAR (+J)

Enter text: (+J) FIG. 26-2 7(i)

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(7) Command: ISOPLANE (..J)

Current isoplane: Top

Enter isometric plane setting [Left/Top/Right] <Right>: RIGHT  
(..J)

Current isoplane: Right

(8) Command: DT (..J)

Current text style: "IS030" Text height: 0.1800 Annotative:  
No

Specify start point of text or [Justify/Style]: J (..J)

Enter an option [Align/Fit/Center/Middle/Right/TL/TC/TR/ML/MC/MR/BL/BC/BR]: BC (..J)

Specify bottom-center point of text: (Select M1Dpoint M2) FIG. 26-2 7(ii)

Specify rotation angle of text <330>: 30 (..J)

Enter text: HOUSE (..J)

Enter text: (..J)

(9) Command: ISOPLANE (..J)

Current isoplane: Right

Enter isometric plane setting [Left/Top/Right] <Top>: FRONT (..J)

Current isoplane: Front

(10) Command: -STYLE (..J)

Enter name of text style or [?] < IS030>: IS0330 (..J)

Specify full font name or font filename (TTF or SHX) < ROMANS.shx>: ROMANTIC (..J)

Specify height of text or [Annotative] <0.2000>: 0.18 (..J)

Specify width factor <1.0000>: (..J)

Specify obliquing angle <30>: 330 (..J)

Display text backwards? [Yes/No] <No>: (..J)

Display text upside-down? [Yes/No] <No>: (..J)

Vertical? [Yes/No] <No>: (..J)

"IS0330" is now the current text style.

(11) Command: DT (..J)

Current text style: "IS0330" Text height: 0.1800 Annotative: No

Specify start point of text or [Justify/Style]: J (..J)

Enter an option [Align/Fit/Center/Middle/Right/TL/TC/TR/ML/MC/MR/BL/BC/BR]: BC (..J)

Specify bottom-center point of text: (Select M1Dpoint M3)

Specify rotation angle of text <30>: 330 (..J)

Enter text: PUBLISHING (..J) FIG. 26-2 7(iii)

Enter text: (..J)

(12) Command: SNAP (..J)

Specify snap spacing or [ON/OFF/Style/Type] <0.5000>: S (..J)

Enter snap grid style [Standard/Isometric] <I>: S (..J)

Specify snap spacing or [Aspect] <0.5000>: (..J)

(13) Command: SNAP (..J)

Specify snap spacing or [ON/OFF/Aspect/Style/Type] <0.5000>: OFF (..J)

(14) Save this File As Module 26-18.DWG

~4

~...

It is a process of generating 3D models for analysis, design, drafting, assembly,

drawing views, realistic look, animation of the assemblies, mass properties, manufacturing and simulation. There are three types of 3D modeling.

(1) 3D Wireframe modelling; (2) 3D Surface modelling (3) 3D Solid modelling.